



TRACKING AND LOCATING SYSTEM ON WEBSITE

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IN COMPUTER ENGINEERING**

**SCHOOL OF INFORMATION TECHNOLOGY
MAE FAH LUANG UNIVERSITY**

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A SENIOR PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

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TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
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2020

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Abstract

This project is made to track various devices. A project that contains 2 parts is Hardware and Software. This project is only an example to develop or study as a guideline of the GPS system. This project is called “TRACKING AND LOCATING SYSTEM ON WEBSITE”, it will work as follows: install the ESP8266 and GPS Module to the device, then the GPS Module will send the value to the web to show the location of the device on the web. Make it possible to know the location of the device.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

GPS Stands for “Global Positioning System” is a satellite navigation system used to determine the ground position of an object. Nowadays, GPS has been developed and applied, such as vehicle tracking, navigation, or personal tracking, etc.

This project is created to track and locate objects in a place. Which is the implementation of the principles of the GPS to work with the IoT. By creating a tracker and locator that will display the value, the coordinates of the device on the website. Which act as an intermediary convert the data received from each GPS Module to be processed for the users to browse, such as the current location of the device. In addition, it is also a development approach to work to meet the needs of various organizations.

1.2 Objective

To develop tracking and locating systems to be applied in the organization.

1.3 Scope

1.3.1 Vehicle tracking device to indicate the location of the vehicle will be displayed on the website.

1.3.2 The system can track and locate devices more than 1 device.

1.4 Expected Result

1.4.1 Able to know the location or coordinates of the device in real-time by display on the website.

1.4.2 As a prototype, for development systems according to the needs of the organization.

1.5 Methodology

- 1.5.1 Proposal preparation
- 1.5.2 Literature review and requirement gathering
- 1.5.3 Presenting the first progress
- 1.5.4 System analysis and design
- 1.5.5 Presenting the progress of the 2nd
- 1.5.6 Prepare for final presentation
- 1.5.7 Pre-project exam
- 1.5.8 System development
- 1.5.9 System testing
- 1.5.10 Prepare for final presentation
- 1.5.11 CE project exam

Task/Month	Computer Engineering Pre-Project				Computer Engineering Project				
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Proposal preparation									
Literature review and requirement gathering									
Presenting the first progress									
System analysis and design									
Presenting the progress of the 2nd									
Prepare for final presentation									
Pre-project exam									
System development									
System testing									
Prepare for final presentation									
CE project exam									

[illegible]

1.7 Resources

1.7.1 Equipment

1.7.1.1 Hardware

- NodeMCU/ESP8266
- GPS Module
- WIFI access point

1.7.1.2 Software

- Visual Studio Code
- Arduino IDE
- NETPIE2020
- Longdo map

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory and Research

A GPS tracking unit or geotracking unit is a navigation device normally carried by a moving vehicle or person or animal that uses the Global Positioning System (GPS) to track the device's movements and determine its geographic position (geotracking). The recorded location data can either be stored within the tracking unit or transmitted to an Internet-connected device using the cellular (GPRS or SMS), radio, or satellite modem embedded in the unit. This allows the location to be displayed against a map backdrop either in real time or when analyzing the track later, using GPS tracking software. Data tracking software is available for smartphones with GPS capability.

2.2 Related Work

2.2.1 GPS and GSM Based Vehicle Tracking System

A program has been developed that is used to locate the exact position of the vehicle and also to navigate the track of the moving vehicle on Google map. The system allows it to track the target anytime and anywhere in any weather conditions. This system is user friendly, easily installable, easily accessible and can be used for various other purposes.

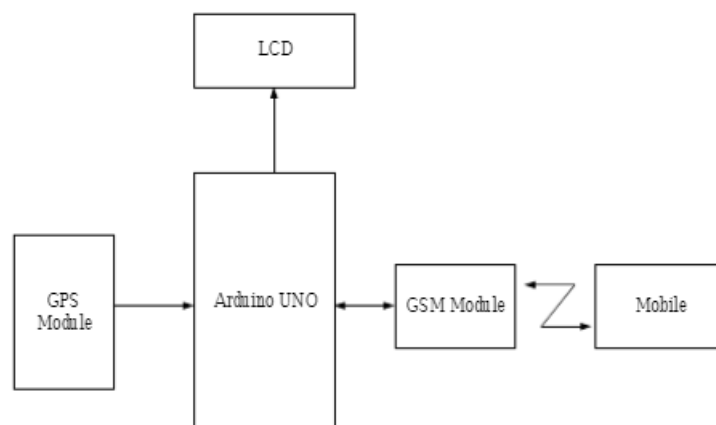


Figure 2.1 Block Diagram of GPS and GSM

2.2.2 The All-Purpose GPS

The idea for this project is simply to be able to create a tracker that is not necessarily incorporated into another communication device, yet is portable. Being a separate device means a longer battery life and more efficient power consumption. It also means multiple applications, especially as a passive GPS. For that reason, my senior project involves building a passive GPS tracker.

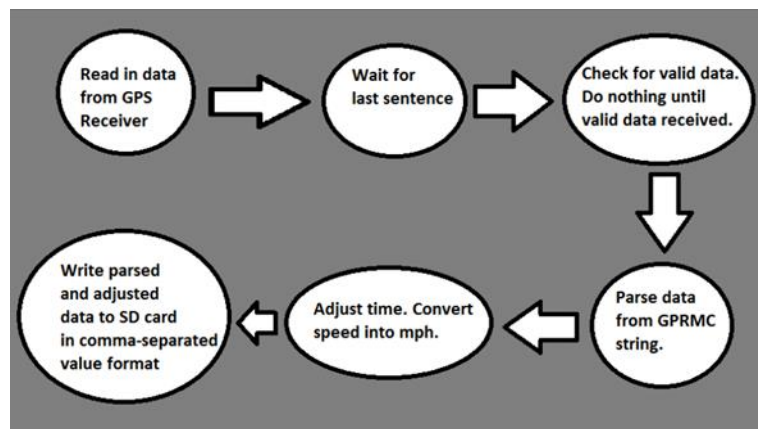


Figure 2.2 Software Pseudocode

The difference is for this project, there will be a website to show the location of the vehicle in particular, using Google map API key to run the map on the website.

- Build a device using the ESP8266 by connecting to the GPS module.
- Create a website page connected with Google API to show the location of the device on Google Map in real time.

2.3 Related Technology

2.3.1 NodeMCU/ESP8266

NodeMCU is an opensource firmware for which opensource prototyping board designs are available. The name "NodeMCU" combines "node" and "MCU" (micro-controller unit). The term "NodeMCU" strictly speaking refers to the firmware rather than the associated development kits. Both the firmware and prototyping board designs are open source.



Figure 2.3 ESP8266

2.3.2 GPS Module (u-blox NEO-6M)

The u-blox NEO-6M global positioning system (GPS) module, a very popular, cost-effective, high-performance GPS module with a ceramic patch antenna, an on-board memory chip, and a backup battery that can be conveniently integrated with a broad range of microcontrollers. Nowadays, two NEO-6M GPS modules are extremely popular — the GY-GPS6MV2 and the GY-GPSV3-NEO. The u-blox NEO-6M GPS engine on these modules is quite a good one, and it also has high sensitivity for indoor applications. Furthermore, there's one MS621FE-compatible rechargeable battery for backup and EEPROM for storing configuration settings. The module works well with a DC input in the 3.3- to 5-V range (thanks to its built-in voltage regulator).



Figure 2.4 GPS Module (u-blox NEO-6M)

2.3.3 Wireless Access point

a wireless access point (WAP), or more generally just access point (AP), is a networking hardware device that allows other Wi-Fi devices to connect to a wired network. The AP usually connects to a router (via a wired network) as a standalone device, but it can also be an integral component of the router itself. An AP is differentiated from a hotspot which is a physical location where Wi-Fi access is available.



Figure 2.5 wireless access point

2.3.4 Visual Studio Code

Visual Studio Code is a distribution of the Code - OSS repository with Microsoft specific customizations released under a traditional Microsoft product license. Visual Studio Code combines the simplicity of a code editor with what developers need for their core edit-build-debug cycle. It provides comprehensive code editing, navigation, and understanding support along with lightweight debugging, a rich extensibility model, and lightweight integration with existing tools and it is updated monthly with new features and bug fixes.



Figure 2.6 Visual Studio

2.3.5 Arduino

The Arduino Integrated Development Environment (IDE) is a cross-platform application (for Windows, macOS, Linux) that is written in functions from C and C++. It is used to write and upload programs to Arduino compatible boards, but also, with the help of third-party cores, other vendor development boards.



Figure 2.7 Arduino

2.3.6 NETPIE2020

NETPIE is an IoT cloud-based platform-as-a-service that helps connect your IoT devices together seamlessly by pushing the complexity from the hands of application developers or device manufacturers to the cloud.



Figure 2.8 NETPIE2020

2.3.7 Longdo Map

Longdo Map Solutions is a suite of discrete services that efficiently manage and manage corporate resources with tired indicators, map replacements, full territories with landmark information, and a wide variety of APIs.

CHAPTER 3

METHODOLOGY

3.1 Block diagram

Real-Time GPS Tracking is a satellite positioning system. The Tracker and GPS Modules will be installed on vehicles. This system will allow us to check the trip data and plan the trip in real-time. Can be done through computers or mobile phones.

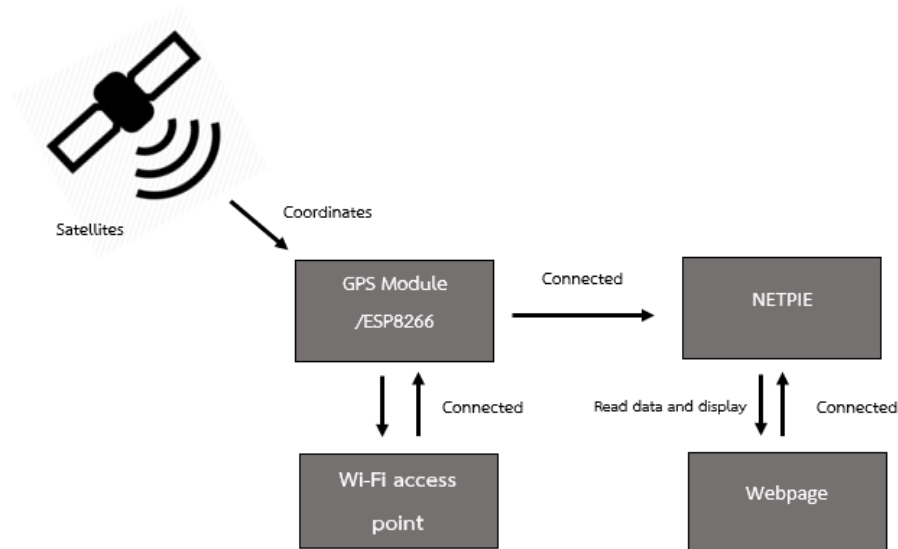


Figure 3.1 Real Time GPS Tracking

For Tracking gem car projects, it is a Tracking gem car project used to track or locate a gem car in real-time because it can check the car position at any time. It will install the GPS module on the gem car. The GPS module will send the latitude and longitude values sent to NETPIE; it will take that value to view the gem car position on the web map.

3.2 Circuit Diagram

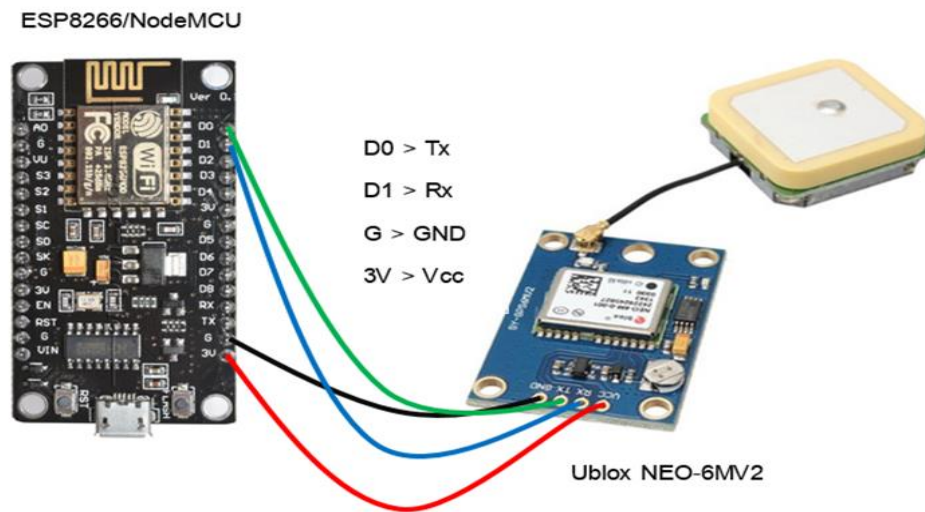


Figure 3.2 GPS Module NEO6MV2 with ESP8266

3.3 Website design

3.3.1 User Interface

This website is made for the convenience of using the gem car service because you can check the position of the gem car.

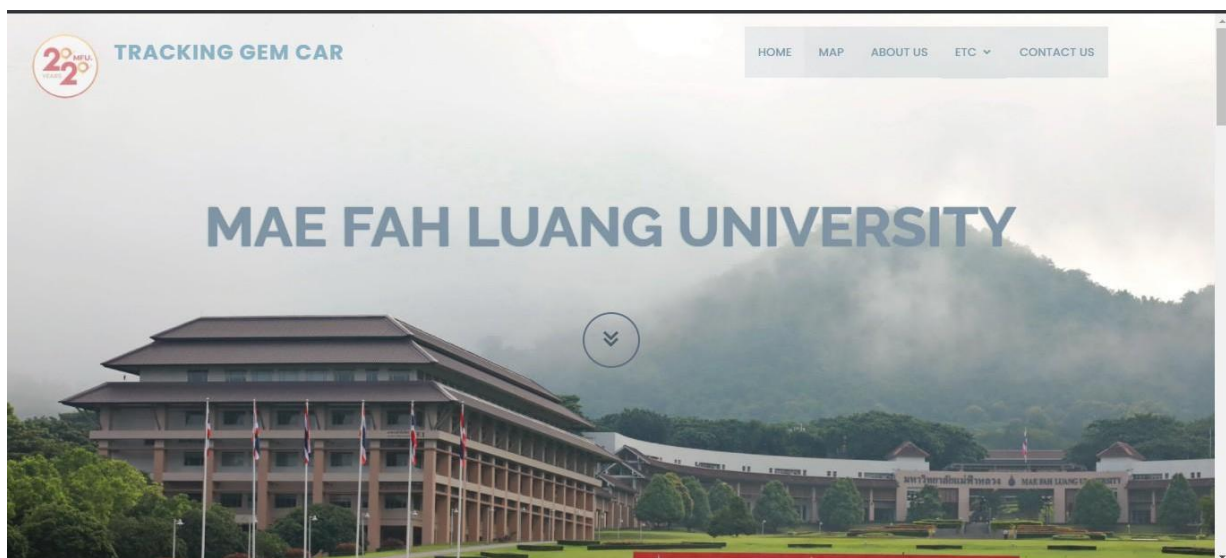


Figure 3.3 First page of website

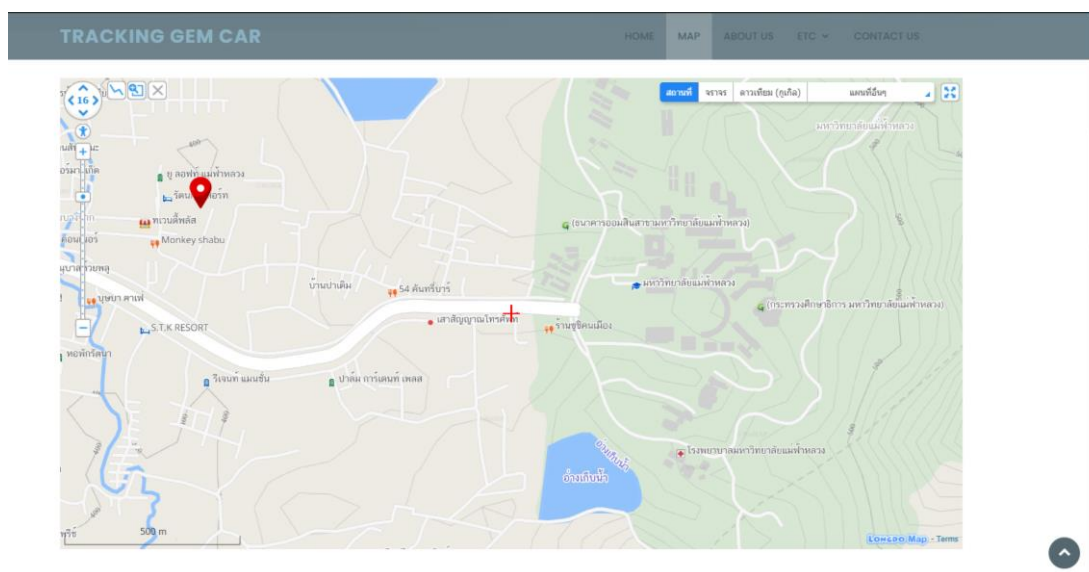


Figure 3.4 Map for show gem car location

3.4 NETPIE2020

3.4.1 NETPIE project page



Figure 3.5 NETPIE page

3.5 Flowchart

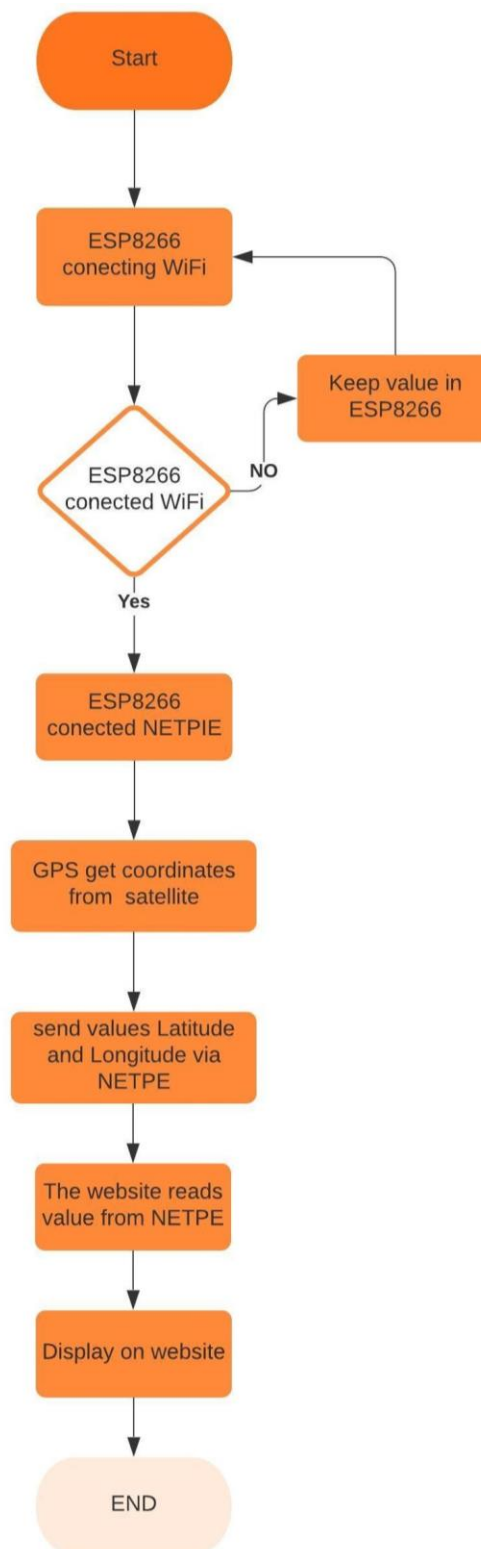


Figure 3.6 Flowchart

CHAPTER 4

RESULTS

In this project, we built a device to track electric vehicles (Gem Car) used on campus, displaying their status on the website we built. Built to track and locate gem cars using GPS with IoT systems.

4.1 Track and display the status on the website

We use the GPS module to display the position and connect it to the ESP8266 by connecting it to NETPIE (IOT platform) connected to the website.

4.2 Device GPS Tracker

In project, device GPS tracker shown in Figure 4.2, is exploited as the IoT processing unit and communication with the website

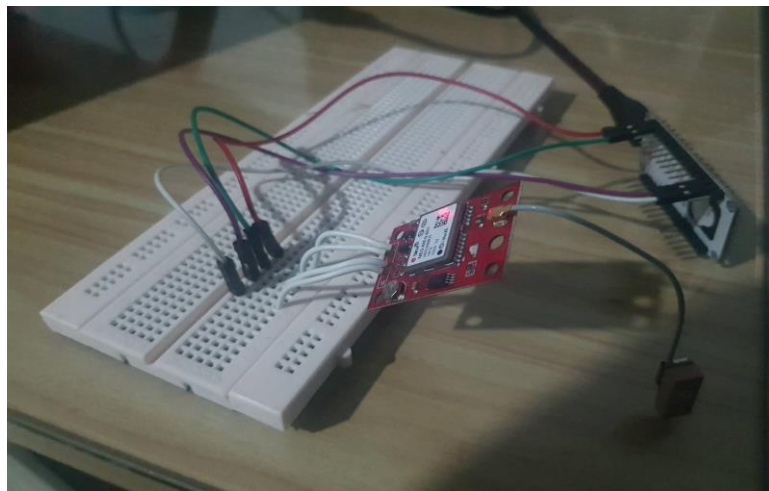


Figure 4.1 Device GPS Track

4.3 Web Server

4.3.1 Web pages

a) On the website will show MAP

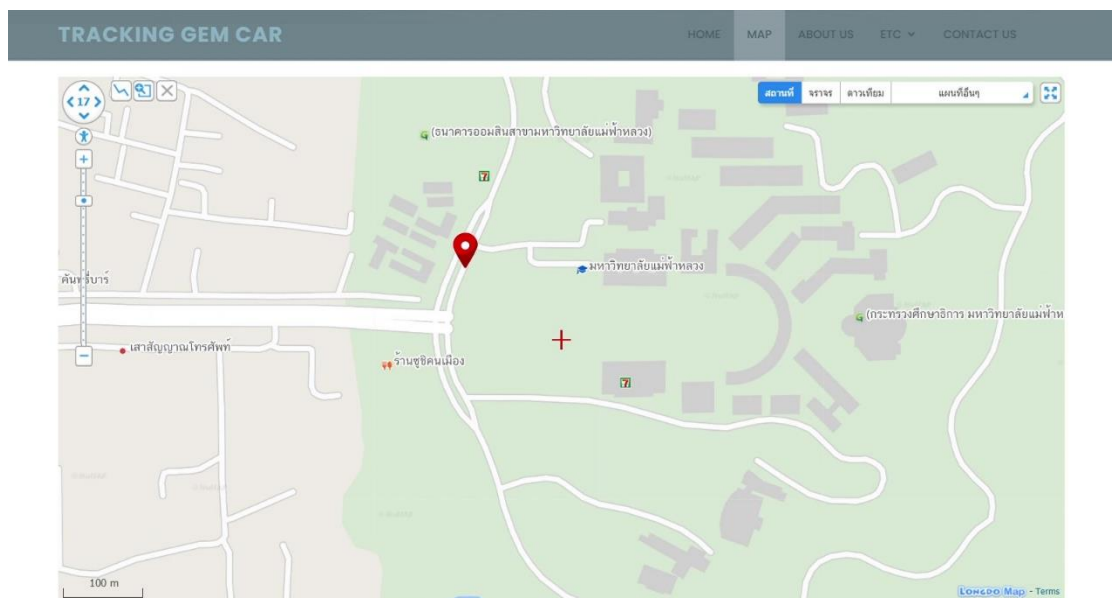


Figure 4.2 status device on map

4.3.2 NETPIE2020

a) To connect ESP8266 and website to NETPIE.

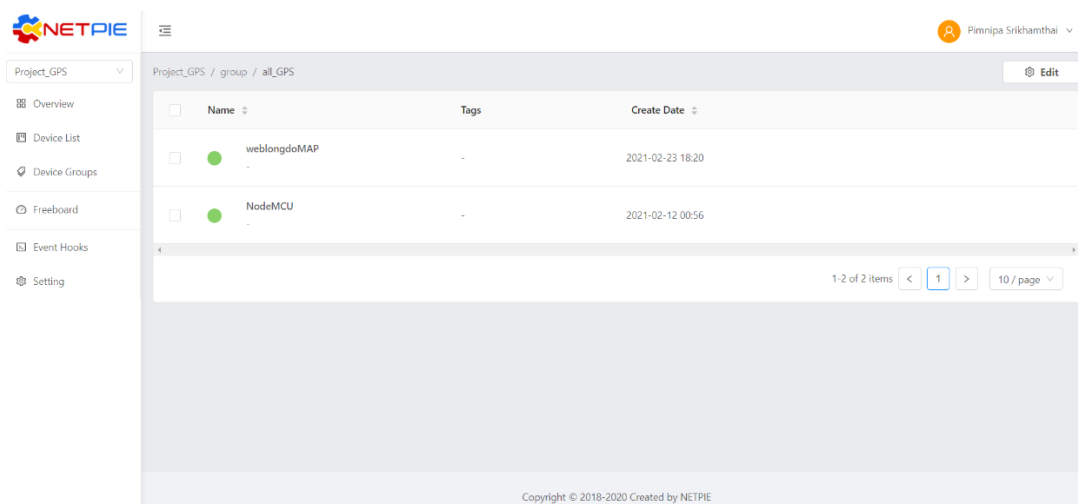


Figure 4.3 NETPIE webpage

4.4 Experiment

4.4.1 Tested by sending a value from the GPS up on the web, it will reset on the web every 5 seconds for location.

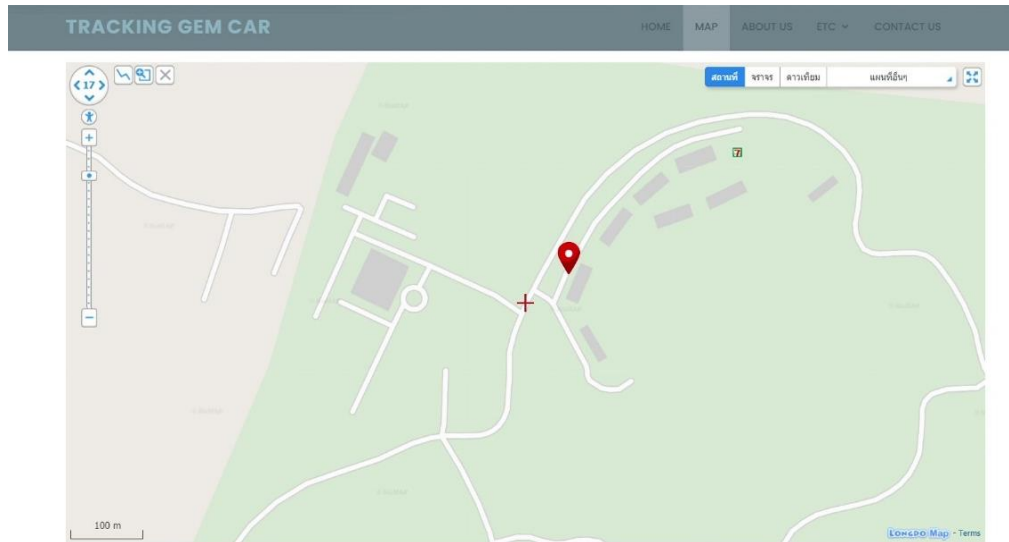


Figure 4.4 Location on the map current web page

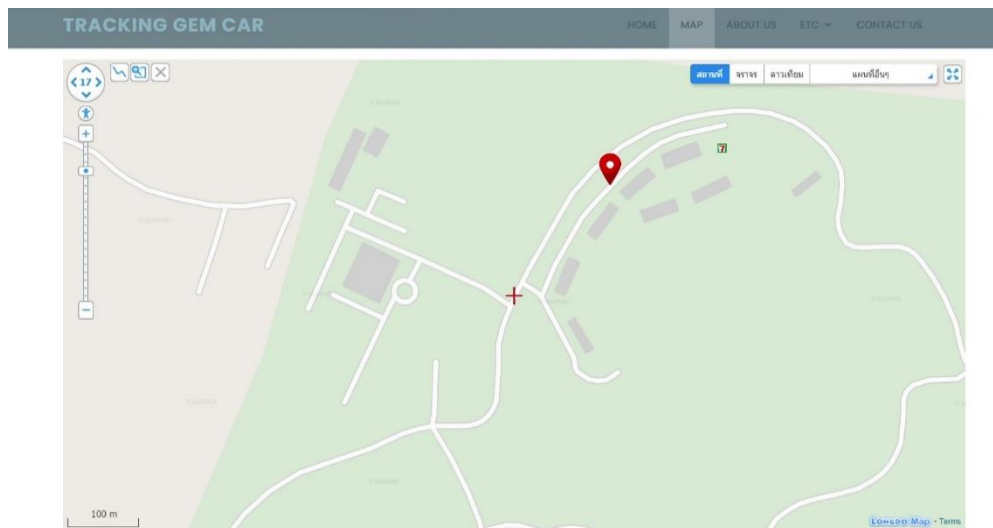


Figure 4.5 Location on map the web page for 5 sec.

4.5 Experiment Result

4.5.1 Tested by sending a value from the GPS up on the web, it will reset on the web every 5 seconds for location.

Round	Latitude	Longitude	Time
1	20.0567147	99.8969159	14.25
2	20.0569522	99.8971817	14.30
3	20.0573868	99.8974285	14.35
4	20.0580889	99.8979746	14.40

Table 1.2 Result for test the system in 5 seconds

After test the system, we can know the location detected by GPS on a webpage and resent every 5 second on the map.

4.5.2 Keep the LOSS value while the device is disconnected

After test to store the LOSS value in the array every 5 seconds so that it can be triggered when the device is disconnected to WIFI.

```

23:38:52.189 -> WiFi connected
23:38:52.189 -> IP address:
23:38:52.189 -> 192.168.43.243
23:38:52.189 -> Attempting MQTT connection...connected
23:38:52.423 -> connected
23:38:52.749 -> 20.047329,99.881050
23:38:58.253 -> 20.047333,99.881050
23:39:03.755 -> 20.047331,99.881050
23:39:09.272 -> 20.047331,99.881050
23:39:14.727 -> 20.047333,99.881035
23:39:19.773 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:20.288 -> 20.047333,99.881035
23:39:25.281 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:25.748 -> 20.047335,99.881042
23:39:30.703 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:31.265 -> 20.047337,99.881042
23:39:36.264 -> Attempting MQTT connection...failed try again in 5 seconds
23:39:36.733 -> 20.047337,99.881050

```

Figure 4.6 Loss value when disconnect is stored every 5 sec.

CHAPTER 5

CONCLUSION AND DISCUSSION

5.1 Conclusion

This project is created to track GEM cars by has created a positioning device for mounting on GEM cars using a GPS module and ESP8266 was created, and a web page for displaying the device's location was created on the page.

5.2 Discussion

Result of working in senior projects. In hardware, GPS module and ESP8266 are connected to Wi-Fi access, which is used to extend the signal for the ESP8266 connection after testing, normal operation. In software, write a program to create a website for receiving value and displaying the position of the device. By connecting esp8266 to netpie2020 and programming part of website pages with netpie2020 as well by creating devices of each part and putting them in the same group, both devices can see the position value that esp8266 sent to netpie2020. Which caused a problem is the internet connection because at some point there is no Wi-Fi signal. For solving that problem will store the LOSS value in the array every 5 seconds when the device is disconnected to WIFI.

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APPENDIX

SOURCE CODE

First, an example of Arduino files.

```
#include <TinyGPS++.h>
#include <ESP8266WiFi.h>
#include <PubSubClient.h>
#include <ESP8266HTTPClient.h>
#include <SoftwareSerial.h>

static const int RXPin = 4, TXPin = 5;
static const uint32_t GPSBaud = 9600;

const char* ssid = "AndroidAP";
const char* password = "09031999";
const char* mqtt_server = "broker.netpie.io";
const int mqtt_port = 1883;
const char* mqtt_Client = "935e6784-1c5d-4b45-aa13-411a32d61aa1";
const char* mqtt_username = "2HhgtL6E1PiY2CKcPZYun9Tfq4G8ZYLk";
const char* mqtt_password = "YHI12ViKzQrhl$_tJHEo!B$vSEJNdnH1";

TinyGPSPlus gps;
SoftwareSerial ss(RXPin, TXPin);

WiFiClient espClient;
PubSubClient client(espClient);

long lastMsg = 0;
int value = 0;
char msg[50];

void setup() {
  Serial.begin(115200);
  Serial.println();
  Serial.print("Connecting to ");
  Serial.println(ssid);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
```

```

}
Serial.println("");
Serial.println("WiFi connected");
Serial.println("IP address: ");
Serial.println(WiFi.localIP());
client.setServer(mqtt_server, mqtt_port);
ss.begin(GPSBaud);
}

```

```

void reconnect() {
  while (!client.connected()) {
    Serial.print("Attempting MQTT connection...");
    if (client.connect(mqtt_Client, mqtt_username, mqtt_password)) {
      Serial.println("connected");
    }
    else {
      int i = 0;
      float var[5];
      char a[50];
      while(i < 3) {
        if (ss.available() > 0){
          gps.encode(ss.read());
          if (gps.location.isUpdated()){
            Serial.print("failed, rc=");
            Serial.print(client.state());
            Serial.println("try again in 5 seconds");
            float latiude = gps.location.lat();
            float longitude = gps.location.lng();
            if(i == 0) {
              var[0] = latiude;
              var[1] = longitude;
              String data1 = String(var[0],6)+"," + String(var[1],6) + " ";
              data1.toCharArray(msg, (data1.length() + 1));
              Serial.println(i);
              Serial.println(data1);
              //box[0] = data.toFloat();
              //Serial.println(data);
            }
            if(i == 1) {
              var[2] = latiude;
              var[3] = longitude;
              String data2 = String(var[2],6)+"," + String(var[3],6) + " ";
              data2.toCharArray(msg, (data2.length() + 1));

```

```

        Serial.println(i);
        Serial.println(data2);
        //box[1] = data.toFloat();
        //Serial.println(data);
    }
    if(i == 2) {
        var[4] = latiude;
        var[5] = longitude;
        String data3 = String(var[4],6)+"," + String(var[5],6) + " ";
        data3.toCharArray(msg, (data3.length() + 1));
        Serial.println(i);
        Serial.println(data3);
        //box[2] = data.toFloat();
        //Serial.println(data);
    }
    i++;
    delay(5000);
}
}
}
}

void loop() {
    if (!client.connected()) {
        reconnect();
        client.publish("@msg/gps", msg);
    }
    client.loop();
    if (ss.available() > 0){
        gps.encode(ss.read());
        if (gps.location.isUpdated()){
            // Latitude
            float latiude = gps.location.lat();
            //Longitude
            float longitude = gps.location.lng();
            String data = String(latiude,6)+"," + String(longitude,6) + " ";
            data.toCharArray(msg, (data.length() + 1));
            Serial.println(data);
            client.publish("@msg/gps", msg);
            delay(1000);
        }
    }
}

```

Second, an example of js files.

```
client = new Paho.MQTT.Client("mqtt.netpie.io", 443, "xxxxxxxxxxxxxxxxxxxxxxxx");
client.onMessageArrived = onMessageArrived;
```

```
var options = {
  useSSL: true,
  userName : "xxxxxxxxxxxxxxxxxxxxxxxx",
  password : "xxxxxxxxxxxxxxxxxxxxxxxx",
  onSuccess: onConnect,
  onFailure:doFail,
}
```

```
client.connect(options);
```

```
function onConnect() {
  client.subscribe("@msg/gps");
}
```

```
function doFail(e){
  console.log(e);
}
```

```
function init() {
  var map = new longdo.Map({
    placeholder: document.getElementById('map')
  });
  //map.location({lon:lng,lat:lati}, true);
  map.location({ lon:99.8942686, lat:20.0447877 }, true);
  map.zoom(17,true)
  //map.location({ lon:98.9582031, lat:18.7951963 }, true);
  //var locationCar = { lon: lng, lat: lati};
  marker = new longdo.Marker({});
```

```
map.Overlays.add(marker);  
}
```

```
var marker=null;  
var lati;  
var lng;  
var a;
```

```
function onMessageArrived(message) {  
  map = new longdo.Map({  
    placeholder: document.getElementById('map')  
  });  
  a = message.payloadString.split(",");  
  console.log(a);  
  
  lati = a[0];  
  lng = a[1];  
  var locat = [  
    ["latlng", Number(lng) , Number(lati) ]  
  ];  
  
  for (var i = 0; i < locat.length; i++) {  
    const loc = locat[i];  
    if(marker){  
      marker = new longdo.Marker({ lon: loc[1], lat: loc[2] });  
      //map.Overlays.add(marker);  
    }  
  }  
  marker = new longdo.Marker({lon:lng , lat: lati});  
  map.Overlays.add(marker);  
}
```